

Divisibility, GCD, LCM & Euclidean Structure

IOQM Grade 9 - Assimilation + First-Step Pack

For the learner who can calculate HCF/LCM but does not yet reliably know when those ideas are the shortest route.

Validated anchors: IOQM-2025-Q02; IOQM-2025-Q27. Exact historical wording remains under source-paper custody.

Concept map

Governing idea: when integers share a divisibility condition, look for structure preserved by subtraction, gcd reduction, or common multiples.

divisibility $\rightarrow$ common divisor $\rightarrow$ differences $\rightarrow$ Euclid / same remainder; divisibility $\rightarrow$ common multiple $\rightarrow$ lcm $\rightarrow$ synchronization.

Decision router

Need a divisor? Think gcd/differences.

Same remainder? Subtract numbers; the divisor divides every difference.

Need the least simultaneous multiple? Think lcm.

Large gcd pair? Use Euclid instead of full factorisation.

Given gcd and lcm of positive integers? Use $\text{gcd}(a,b) \cdot \text{lcm}(a,b) = ab$ before casework.

Reconnect

Greatest divisor leaving the same remainder on 173 and 239: first move = subtract.

Least positive integer divisible by 12, 15 and 18: this is a common-multiple problem.

To shrink $\text{gcd}(987,610)$, preserve the gcd while replacing the larger number by a remainder.

Make sense

If $d|a$ and $d|b$, then $d|(a-b)$. Conversely, if $d|b$ and $d|(a-b)$, then $d|a$. Therefore $\text{gcd}(a,b) = \text{gcd}(b, a - qb)$. If $a = dq + r$ and $b = dp + r$, then $d|(a-b)$: equal remainder becomes a difference problem.

Try before hints

A. Greatest integer dividing 221, 323 and 425 with the same remainder. H1: subtract. H2: use differences 102 and 102. H3: answer 102.

B. Least positive integer divisible by 18,24,30. H1: divisor or multiple? H2: lcm. H3: answer 360.

First-Step Reference

same remainder $\rightarrow$ subtract $\rightarrow$ gcd of differences

least positive integer divisible by several numbers $\rightarrow$ lcm

large gcd pair $\rightarrow$ Euclidean algorithm

gcd and lcm both given $\rightarrow$ product invariant + normalization

common divisor of linear combinations $\rightarrow$ form useful integer combinations

Recognition lab

1) same remainder for 188,296: write the first divisibility line.

2) least multiple of 14,20,35: write the first operation.

3) $\text{gcd}(12345,5432)$: write the first Euclidean reduction.

4) $\text{gcd}=9, \text{lcm}=252$: write an invariant involving ab .

H0 mastery

1) Greatest divisor leaving equal remainder for 428,596,764.

2) Least $N > 11$ leaving remainder 11 on division by 18,24,30.

3) Compute $\gcd(123456, 7890)$ without full prime factorisation.

4) $\gcd(a, b) = 14$ and $\text{lcm}(a, b) = 840$: find ab .

5) If d divides $11n+8$ and $7n+5$, find the largest possible d .

Adopt: ask first whether the problem wants a divisor, a multiple, or an invariant preserved by subtraction.